CHAPTER 4 - SYSTEM DESIGN & DEVELOPMENT

4.1 Introduction

This chapter shows how we turned Chapter 3 requirements into a working design and what we built in the development sprint. TrackWIL replaces the PDF and email logbook with a small web-app. Students log workplace hours, the employer checks and entry through a link and does not need an account to do so. The WIL coordinator can see who is on track, who is waiting on a mentor, and who is overdue.

For week 3 we drew the architecture, UML, and database design. We built the app at the same time. The code in `05_Source_Control/trackwil` covers the three core features from the proposal.

4.2 System Architecture

TrackWIL is a Next.js application. The browser talks to a single Node.js process. That process serves the React pages, runs the business logic, and reads and writes to the JSON database. Uploaded evidence sits on disk.

Main components:

Component	**Responsibility**
Student area	Dashboard, new entries, evidence, sharing
Coordinator area	Table of students with status, verifying hours
Public verify	Mentor entry approval
Auth	Login, logout, signup
Verify action	Checks token, updates entry status

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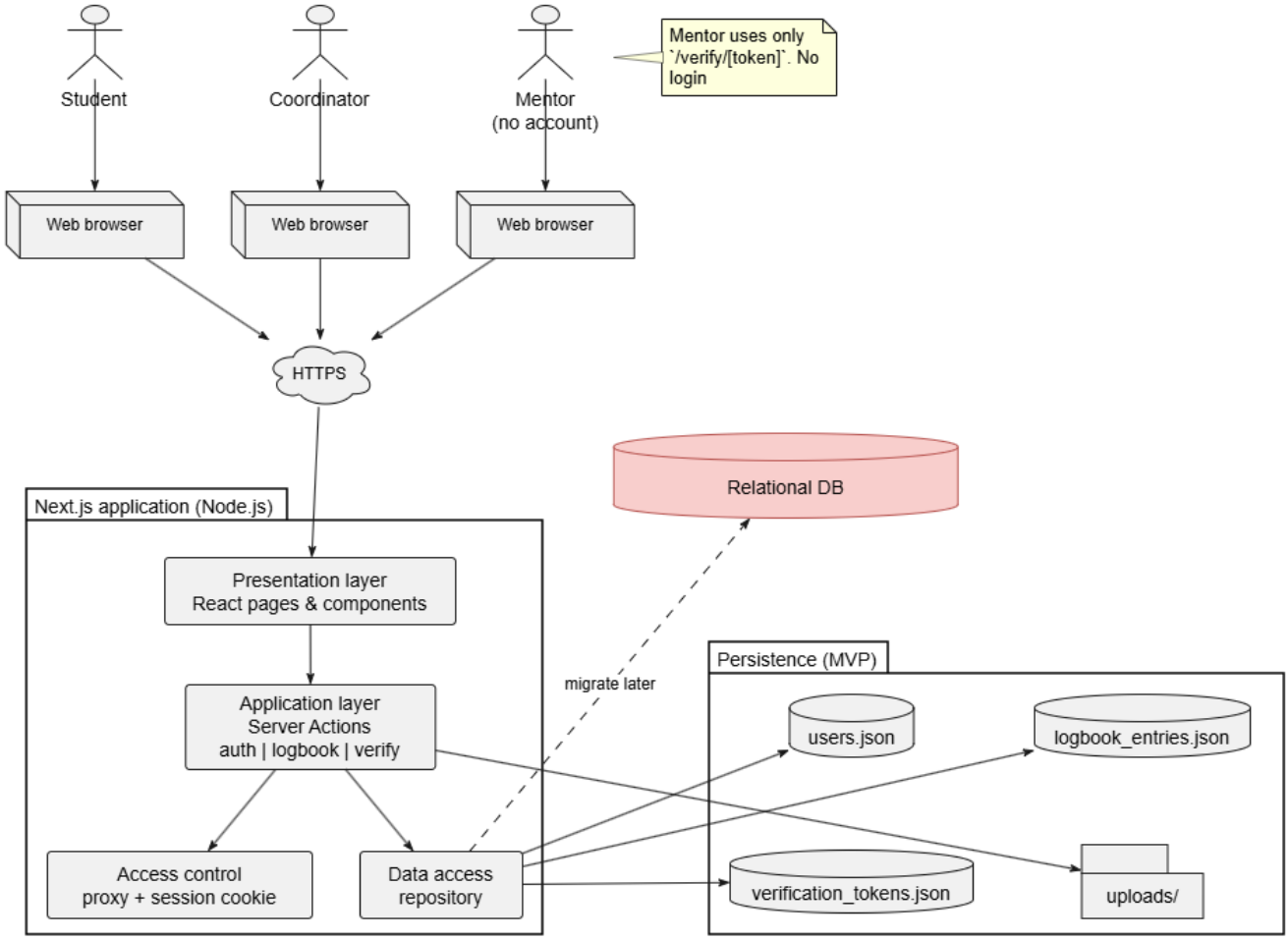


Figure 1 Architecture Diagram

4.3 UML Design

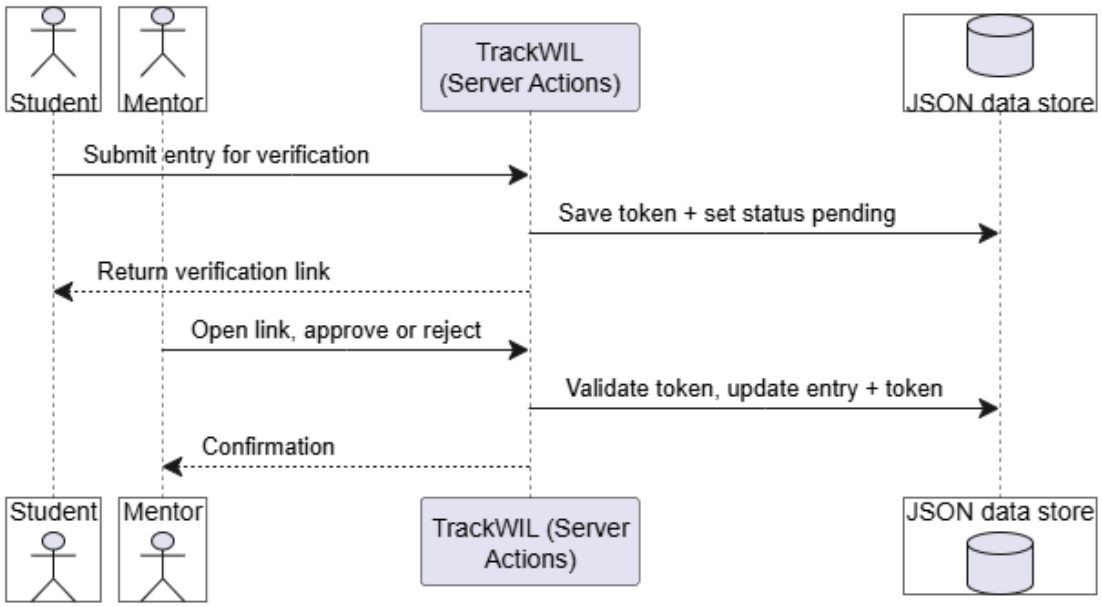


Figure 2 Sequence Diagram

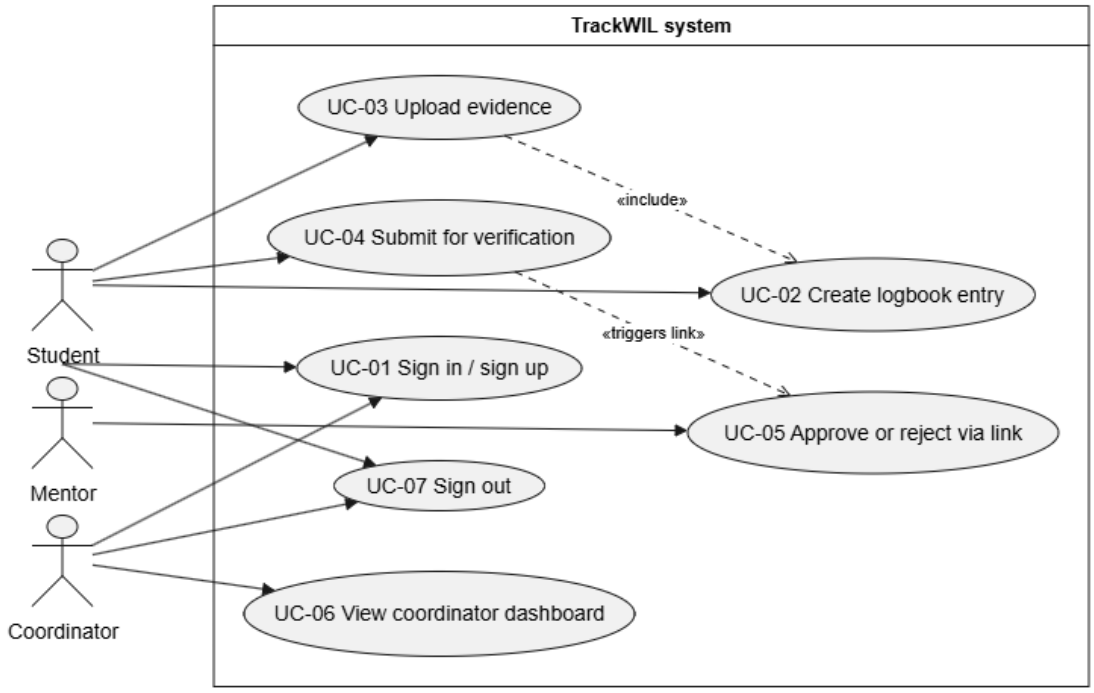


Figure 3 Use Case Diagram

The sequence diagram shows mentor verification as message between the student, the mentor, TrackWIL, and the database.

The use-case diagram has 3 actors:

- Student – signs up, logs hours, uploads evidence, and shares the entry for checking.
- Mentor – Approves/rejects a student's entry.
- Coordinator – Monitors progress across students.

4.4 Database Design

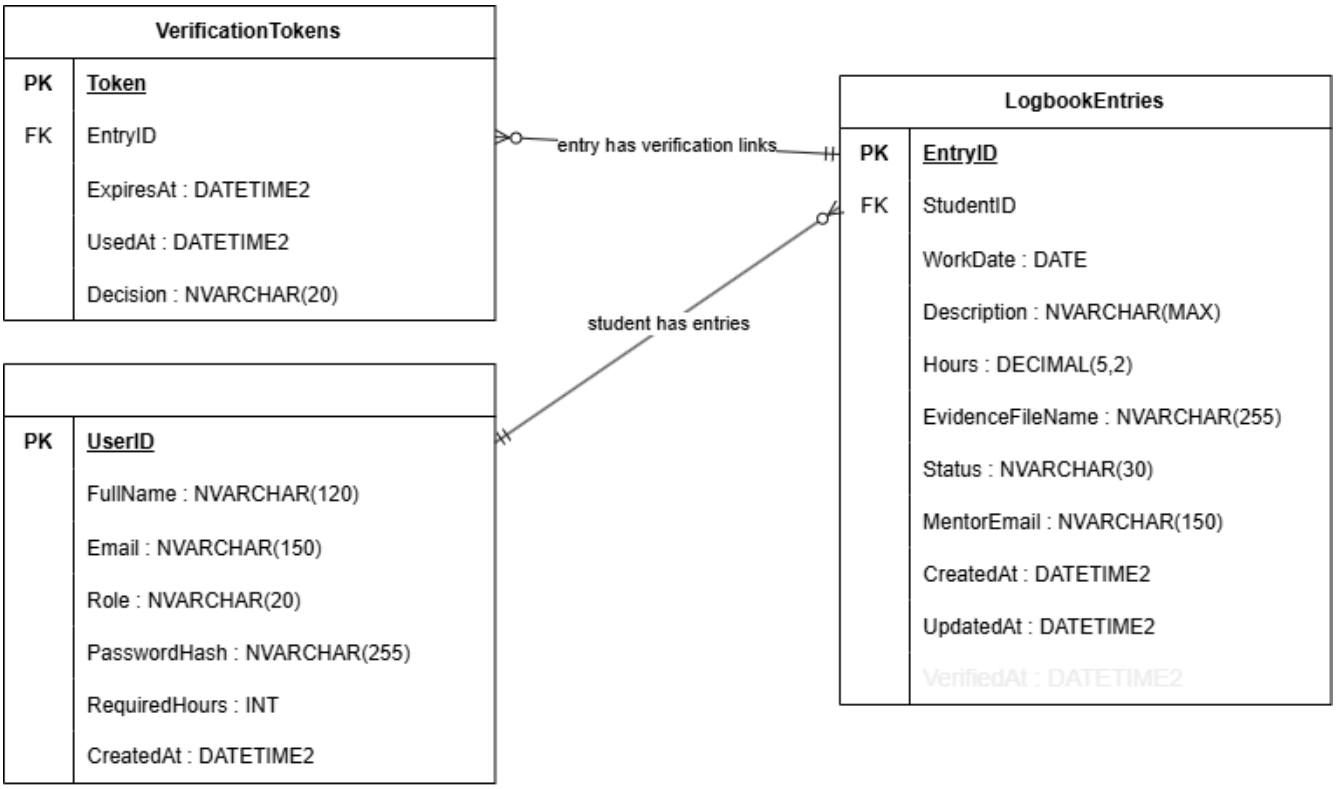


Figure 4 ERD

4.5 User Interface Design

Show wireframes or prototypes and explain usability decisions.

4.6 Development Environment

- Languages: TypeScript, HTML and CSS through JSX, SQL in the schema script
- Framework: Next.js 16
- IDE: Visual Studio Code
- Database: JSON file system
- Libraries: Tailwind CSS
- Cloud services: none. Local only
- Version control: Git and GitHub (https://github.com/Noobey32/TrackWIL)